

- 29** In the absence of any radioactive source, the average value of the background radiation is found to be 30 counts per second.

When a radioactive source with half-life of 5 hours is placed at a fixed distance from the counter, the average total count rate is 210 counts per second.

What average count rate is expected with the source in the same position 10 hours later?

- A** 45 s^{-1}
- B** 53 s^{-1}
- C** 60 s^{-1}
- D** 75 s^{-1}

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30 Which isotope decays by emission of a β -particle to produce ^{111}Cd ?